

BugTracker — OpenShift Runbook

Complete Deployment & Operations Guide

Project: **BugTracker v2.1.0**

Platform: **OpenShift Developer Sandbox**

Namespace: **the-noid-dev**

GitHub: **<https://github.com/THENOID-404/Bug-Tracker>**

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1. Prerequisites

Before deploying, make sure you have these ready:

Requirement	Where to Get	Status
GitHub Account	github.com	Sign up free
OpenShift Developer Sandbox	developers.redhat.com/developer-sandbox	Sign up free
oc CLI tool	mirror.openshift.com/pub/openshift-v4/clients/oc	Download & install
Node.js (optional)	nodejs.org	Only for local testing

2. Project Files Overview

File	Purpose	Edit?
openshift-buildconfig.yaml	Tells OpenShift how to build from GitHub	YES
openshift-deployment.yaml	Tells OpenShift how to run the app	YES
Dockerfile	Instructions to build the Docker image	Sometimes
nginx.conf	Web server configuration	No
docker-compose.yaml	For local development only — NOT used by OpenShift	No
package.json	Node.js dependencies list	No
src/	React source code files	Yes (your app code)
public/	Static assets	Yes (your app assets)

3. One-Time Setup

Step 1 — Install oc CLI

Download from: mirror.openshift.com/pub/openshift-v4/clients/oc

After installing, verify it works:

```
oc version
```

Step 2 — Login to OpenShift

Go to your Developer Sandbox dashboard → Click '**Copy login command**' → Paste in CMD:

```
oc login --token=YOUR_TOKEN --server=YOUR_SERVER_URL
```

** Token expires every 24 hours. Repeat this step daily.*

Step 3 — Verify you are in correct project

```
oc project
```

You should see: **the-noid-dev**

If not, switch to it:

```
oc project the-noid-dev
```

4. File Changes Required

openshift-buildconfig.yaml — 2 changes

Change 1: Your GitHub repo URL

```
uri: https://github.com/THENOID-404/Bug-Tracker.git
```

Change 2: Webhook secret (anything simple)

```
secret: abc123
```

openshift-deployment.yaml — 1 change

Change 1: Image registry path with your project name

```
image: image-registry.openshift-image-registry.svc:5000/the-noid-dev/bugtracker:latest
```

5. First Deployment (Step by Step)

Step 1 — Open CMD and go to project folder

```
cd C:\Users\THE NOID\Downloads\bugtracker\bugtracker
```

Step 2 — Login to OpenShift (get token from sandbox dashboard)

```
oc login --token=YOUR_TOKEN --server=YOUR_SERVER
```

Step 3 — Apply buildconfig file

```
oc apply -f openshift-buildconfig.yaml
```

Step 4 — Apply deployment file

```
oc apply -f openshift-deployment.yaml
```

Step 5 — Start the build (watch it live)

```
oc start-build bugtracker --follow
```

Step 6 — Check pods are running

```
oc get pods
```

Step 7 — Get your live website URL

```
oc get route
```

** The last command gives you a URL like: bugtracker-the-noid-dev.apps.sandbox.openshiftapps.com*

6. Every Code Change Flow

Every time you change code and push to GitHub, follow this flow:

Step	Action	Command
1	Change code on GitHub	Edit file → Commit changes
2	Trigger new build	oc start-build bugtracker --follow
3	Restart deployment	oc rollout restart deployment/bugtracker
4	Hard refresh browser	Ctrl + Shift + R

** oc rollout restart is only needed if changes are not visible after hard refresh.*

7. Health Check Commands

Command	What it does
oc get pods	See all pods and their status
oc get pods -w	Watch pods live (auto-refreshes)
oc get deployment bugtracker	Check deployment health
oc get route	Get your live website URL
oc adm top pods	See CPU and Memory usage
oc logs -f POD_NAME	Watch live app logs
oc logs --tail=50 POD_NAME	See last 50 lines of logs
oc describe pod POD_NAME	Full detailed pod information

Health Check Script (healthcheck.bat)

Create this file on your Desktop and double-click anytime for a full report:

```
@echo off
echo =====
echo    BUGTRACKER HEALTH CHECK
echo =====
echo.
echo --- PODS STATUS ---
oc get pods
echo.
echo --- DEPLOYMENT STATUS ---
oc get deployment bugtracker
echo.
echo --- ROUTE URL ---
oc get route
echo.
echo --- RESOURCE USAGE ---
oc adm top pods
pause
```

8. Troubleshooting Guide

Issue: Build fails with wrong GitHub URL

Error: error: failed to fetch repository "https://github.com/yourorg/..."

Fix:

```
oc patch bc/bugtracker -p '{"spec":{"source":{"git":{"uri":"https://github.com/THENOID-404/Bug-Tracker.git"}}}}'
```

Issue: Build fails: npm ci error

Error: npm error: package-lock.json missing

Fix:

```
Change 'npm ci' to 'npm install --legacy-peer-deps' in Dockerfile
```

Issue: Build fails: ajv module not found

Error: Error: Cannot find module 'ajv/dist/compile/codegen'

Fix:

```
In Dockerfile: RUN npm install --legacy-peer-deps && npm install ajv@^8.11.0 ajv-keywords@^5.0.0
```

Issue: Pods show ImagePullBackOff

Error: 0/1 ImagePullBackOff

Fix:

```
oc set image deployment/bugtracker bugtracker=image-registry.openshift-image-registry.svc:5000/the-noid-dev/bugtracker:latest
```

Issue: Code changes not showing

Error: App shows old content after build

Fix:

```
oc rollout restart deployment/bugtracker then Ctrl+Shift+R in browser
```

Issue: Wrong project / namespace

Error: Commands affecting wrong project

Fix:

```
oc project the-noid-dev
```

9. Crash & Recovery Operations

Intentionally Crash the App

Scale down to 0 pods (kills the app):

```
oc scale deployment bugtracker --replicas=0
```

Recover the App

Scale back up to 2 pods:

```
oc scale deployment bugtracker --replicas=2
```

Watch Recovery in Real Time

Open 2 CMD windows side by side:

Window 1 — Watch live:

```
oc get pods -w
```

Window 2 — Delete one pod:

```
oc delete pod bugtracker-55b74db9f7-dbm1z
```

** OpenShift automatically creates a replacement pod in seconds!*

Check Crash Logs

See why a pod crashed (even after it restarts):

```
oc logs POD_NAME --previous
```

Full crash details:

```
oc describe pod POD_NAME
```

Look for the **Events** section at the bottom of describe output.

Where Are Issues Stored?

Pod Type	Stores
build pod (bugtracker-X-build)	Build logs: npm install, compile errors, Dockerfile errors
running pod (bugtracker-XXXXX)	Runtime logs: crashes, errors, live requests

10. Pod Reference Guide

Pod Name Pattern	Type	Purpose	Delete?
bugtracker-XXXXX-YYYYY (Running)	Live Server	Serves your website to users. OpenShift runs 2 for high availability.	NEVER
bugtracker-N-build (Completed)	Build Log	Records how the app was built. Latest one useful for debugging.	Safe (keep latest)
bugtracker-N-build (Error/Init:Error)	Failed Build	A build attempt that failed.	Safe to delete

Delete Failed Build Pods

```
oc delete pods --field-selector=status.phase=Failed
```

Delete Specific Build Pods

```
oc delete pod bugtracker-2-build bugtracker-3-build bugtracker-4-build
```

11. Quick Reference Card

Task	Command
Login to OpenShift	oc login --token=TOKEN --server=URL
Switch project	oc project the-noid-dev
Apply config files	oc apply -f openshift-buildconfig.yaml
Start build	oc start-build bugtracker --follow
Restart deployment	oc rollout restart deployment/bugtracker
Get live URL	oc get route
See all pods	oc get pods
Watch pods live	oc get pods -w
See live logs	oc logs -f POD_NAME
See crash logs	oc logs POD_NAME --previous
Pod details	oc describe pod POD_NAME
CPU/Memory usage	oc adm top pods
Crash app	oc scale deployment bugtracker --replicas=0
Recover app	oc scale deployment bugtracker --replicas=2
Delete failed pods	oc delete pods --field-selector=status.phase=Failed
Fix image path	oc set image deployment/bugtracker bugtracker=image-registry.opensh

12. How This App Was Deployed — Full Story

This section documents exactly what tools were used, what prerequisites were followed, and every step taken to deploy BugTracker from zero to live on OpenShift.

Tools & Technologies Used

Tool	Purpose	Where
OpenShift Developer Sandbox	Free cloud platform to host the app	developers.redhat.com/developer-sandbox
oc CLI	Command line tool to talk to OpenShift	Installed on Windows PC
GitHub	Store source code, trigger builds	github.com/THENOID-404/Bug-Tracker
Docker / Dockerfile	Package the app into a container image	Already in project files
nginx	Web server to serve the React app	Inside Docker container
Node.js 18 (Alpine)	Build the React app inside Docker	Used during build only
React + react-scripts	Frontend framework for the app	package.json dependencies
Windows CMD	Run all oc commands	Local machine

Prerequisites Followed Before Deployment

#	Prerequisite	What Was Done
1	GitHub Account	Created account, uploaded all project files to new repo: THENOID-404/Bug-Tracker
2	OpenShift Sandbox	Signed up at developers.redhat.com, project 'the-noid-dev' was auto-created
3	oc CLI installed	Downloaded and installed on Windows. Verified with 'oc version'
4	Login token	Copied login command from Sandbox dashboard, ran 'oc login' in CMD
5	Project verified	Confirmed active project is 'the-noid-dev' using 'oc project'
6	YAML files reviewed	Checked openshift-buildconfig.yaml and openshift-deployment.yaml for required changes

File Changes Made Before First Deploy

File	Line Changed	From	To
openshift-buildconfig.yaml	uri:	https://github.com/yourorg/bugtracker.git	https://github.com/THENOID-404/Bug-Tracker.git
openshift-buildconfig.yaml	secret:	your-github-webhook-secret	abc123
openshift-deployment.yaml	image:	bugtracker:latest	image-registry.openshift-image-registry.svc:5000/bugtracker:latest
Dockerfile	RUN npm ci	npm ci --legacy-peer-deps	npm install --legacy-peer-deps && npm install

Issues Faced & How They Were Resolved

Build 2-4 — Init:Error	Problem: Wrong GitHub URL in buildconfig — still had placeholder 'yourorg/bugtracker.git' Solution: Used 'oc patch' command to directly update the URL in OpenShift without editing file again
Build 5 — Error	Problem: npm ci failed — no package-lock.json file existed in the repo Solution: Changed 'npm ci' to 'npm install' in Dockerfile on GitHub

Build 6 — Error	Problem: ajv module not found — version conflict between react-scripts 5.0.1 and Node 18 Solution: Added 'npm install ajv@^8.11.0 ajv-keywords@^5.0.0' to Dockerfile install command
Build 7 — Completed	Problem: Build succeeded but pods showed ImagePullBackOff Solution: Used 'oc set image' to set the correct internal registry path for the image
After build 10 — Running	Problem: Code changes not reflecting in browser after rebuild Solution: Used 'oc rollout restart deployment/bugtracker' + hard refresh (Ctrl+Shift+R)

Final Deployment Architecture

After successful deployment, the running setup looks like this:

Component	Details
GitHub Repo	THENOID-404/Bug-Tracker — source of truth for all code
OpenShift BuildConfig	Pulls code from GitHub, builds Docker image using Dockerfile
ImageStream	Stores the built Docker image internally in OpenShift registry
Deployment	Runs 2 replica pods from the built image for high availability
Service	Internal load balancer — routes traffic between the 2 pods on port 8080
Route	Public HTTPS URL — exposes the app to the internet with TLS/SSL
Live Pods	2 x Running pods (bugtracker-55b74db9f7-*) serving the React app via nginx